Safran T4 i3000 Extension Plan

Version 0.1.0 / November 3, 2026

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Scope and evidence

This planning phase began only after MHM2010 Correction Workbench 0.1.0 was posted for download. No T4 implementation, live query, instrument adjustment or remote repository push was performed.

The reviewed repository's T4 examples use IPv4 UDP, example port 14000, the command `MONIT;` followed by CRLF, and a dollar-prefixed 113-hex-character payload [1]. Safran's datasheet describes Ethernet monitoring and remote functions, but not the complete MONIT protocol or the synthesizer/drift control transfer equations [2]. The AB monitor source supplies useful qtpy and pyqtgraph patterns for bounded live previews and in-place updates [3].

The reviewed commit is `9682a2ffbcd0b0df6612d4aef3619d7bdaf6cc66`. The T4 source refers to page 47 of an operation manual, but that manual has not been independently obtained and checked [1].

Boundary: do not reuse the MHM2010 DDS constants for T4. Reuse timing fits, stability, provenance and export methods, while keeping the T4 actuator mapping disabled until the OEM relationship is verified.

Source review findings

These are static-code findings, not observed failures on the physical maser [1].

Area	Finding and planned remedy
Retries	A timeout closes the socket, then the loop attempts to reuse it. Use an explicit socket lifecycle and fault-injected retry tests.
Response typing	Bytes/None are returned despite an optional-string annotation; the caller decodes before checking failure. Use structured results.
Peer identity	The received address is not verified. Validate the intended peer and permitted reply port; isolate late and unrelated datagrams.
Parser	Empty input, loose delimiters and ad hoc stripping require replacement with strict byte framing and typed exceptions.
GUI/process exit	Decoder errors invoke Tk and exit. Keep the new parser pure; qtpy handles errors at the application boundary.
Status 39/40	An analog scale is applied to a lock-labeled byte; a one-bit comment conflicts with whole-nibble decoding. Preserve raw fields pending manual and known-state captures.
Unavailable channels	NaN-gain channels are omitted. Retain identity and raw code with an unavailable flag and null engineering value.
Import behavior	Companion examples query at import time. New modules must be inert on import and contain no fixed site endpoint.

[1] T4 communication source: https://raw.githubusercontent.com/Braveheart53/GBO_Maser_Control/main/T4_Maser_Comms.py

[2] Safran datasheet: https://safran-navigation-timing.com/wp-content/uploads/2024/03/SAFRAN_IMASER-3000_Datasheet_3-11-2024_US.pdf

[3] GBO control repository: https://github.com/Braveheart53/GBO_Maser_Control/tree/main

Proposed architecture and staged files

- **Pure decoder**: bytes plus a versioned schema produce raw codes, engineering values and quality flags. No network, file or GUI side effects.
- **Read-only acquisition**: a monitoring-only allowlist, request/receive UTC, monotonic round-trip timing, attempt count, raw frame and parse status.
- **Port ownership**: start with one owned listener and serialized queries; do not assume ephemeral local ports work. Multiple instruments must not compete to bind the same local port.
- **Persistence**: raw frames precede scaled data. Include schema hash, instrument identity, firmware, sequence and unavailable-channel masks.
- **GUI**: separate T4 tab, qtpy/PySide6 and pyqtgraph, in-place curves, bounded previews, linked axes and crosshair/value displays, following the AB exemplar [3].
- **Timing**: explicitly map a genuine T4/reference FITS pair. Do not assume a SigmaTau or MicroSem label identifies the T4.
- **Corrections**: a separate OEM-calibrated adapter, initially disabled. No state-changing command path in the first acquisition release.

The staged `t4_channel_schema_draft_v0.1.0.json` records all 41 source-channel identities and provisional gains, without a host address or personal path. Status fields and unavailable channels are explicitly marked. It is not an approved calibration table.

Milestones and acceptance gates

Milestone	Required gate
Protocol evidence	Actual manual, firmware identity, raw captures and OEM display comparisons resolve frame layout, scaling and status ambiguity.
Offline parser	Captured valid frames agree exactly with the OEM display; empty, truncated, oversized, bad-hex and wrong-terminator frames fail safely.
Read-only poller	Timeout/success, exhausted retries, wrong peer, late reply, duplicate, occupied port, cancellation and restart tests pass.
Telemetry GUI	Bounded memory, zoom preservation, stale-state indication, worker cancellation and successful read-only site test.
Timing integration	Verified T4 channel identity/sign, reference and telemetry overlap; gaps preserved; MHM2010 regressions remain unchanged.
Correction advisory	OEM transfer gain, units, existing ramp, precision and epoch verified with independent synthetic tests.

Persistence tests must include disk failure, interrupted append and restart. Timing tests must include host-clock steps, timezone/DST display behavior and monotonic round-trip measurement. Host receive timestamps are not precision timestamps generated by the maser.

[3] GBO control repository: https://github.com/Braveheart53/GBO_Maser_Control/tree/main

Evidence needed before implementation

- **Manual**: operation/remote-command manual, the channel table referenced as page 47, lock-bit definitions and frequency/drift control sections.
- **Identity**: installed firmware/software and screenshots of current field names, units, values and enabled modes.
- **Telemetry**: unedited MONIT replies with matching displays, including known lock states and installed-option differences if available.
- **Timing**: actual FITS names for the T4 and its reference, with an overlapping recording.
- **Access**: connected bracca for the authorized staging directory; separate explicit approval of a read-only endpoint before any network test.

No timeline or credit estimate is assumed. These evidence and acceptance gates govern readiness.

Appendix: Proposed equations

The repository's provisional analog conversion and payload accounting are [1]:

$$v_i = g_i c_i, \qquad L_{\text{hex}} = 32(3) + 8(2) + 1 = 113.$$

Neither the gains nor status semantics become OEM-verified merely because they appear in the source.

For a future actuator variable u, define a locally verified fractional gain G. A small-signal advisory model could then use:

$$G = \frac{\partial \ln f_{\text{out}}}{\partial u}, \qquad \Delta u \simeq -\frac{y}{G}, \qquad \Delta \dot{u} \simeq -\frac{r}{G}, \qquad r = \frac{dy}{d(\text{day})}.$$

These are generic local relationships, not T4 firmware equations. No numerical gain is assigned. Nonlinear gain, existing steering, discrete updates and saturation require the actual T4 model.

IEEE reference table

Ref.	Bibliographic entry	DOI or URL
[1]	W. W. Wallace, "T4_Maser_Comms.py," GBO_Maser_Control, version comments through 1.0.1, last-edit header Jun. 5, 2025. Reviewed commit `9682a2ffbcd0b0df6612d4aef3619d7bdaf6cc66`. Accessed Nov. 3, 2026.	https://raw.githubusercontent.com/Braveheart53/GBO_Maser_Control/main/T4_Maser_Comms.py
[2]	Safran, "iMaser 3000," product datasheet, Mar. 11, 2024. Accessed Nov. 3, 2026.	https://safran-navigation-timing.com/wp-content/uploads/2024/03/SAFRAN_IMASER-3000_Datasheet_3-11-2024_US.pdf
[3]	W. W. Wallace, "GBO_Maser_Control," including AB Power Meter Monitor 1.7.0 and `ab_live_plots.py` 0.1.0, GitHub. Reviewed commit as above. Accessed Nov. 3, 2026.	https://github.com/Braveheart53/GBO_Maser_Control/tree/main